CONTINUOUS INTERNAL ASSESSMENT - 1

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Course Name ⇒ Computer Networks

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① · OSI Model and Transport layer Services:

(a) Functions of Data link layer and Transport Layer:

1. Data link layer:

- Framing: Divides data into frames
- MAC Addressing: Identifies devices physically
- Error Detection: Detects error using CRC.
- Flow Control: Control data transmission
- Medium Access Control: Manages access to the medium.

2. Transport Layer:

- Provides Process - to - Process delivery.
- Segmentation - Divides data to Smaller segments
- uses port numbers for communication
- Provides flow and congestion control.
- Uses ACK and retransmission for error recovery.

How both layers work together:

- Data link layer provides hop-to-hop reliable.
- Transport layer provides end-to-end reliability
- Both help in reliable data delivery.

Diagram:

Application layer
↓
Transport layer
↓
Network layer
↓
Data link layer
↓
Physical layer

(b) Difference Between TCP and UDP

TCP	UDP
* connection-oriented	Connectionless
* Reliable delivery	Best-effort delivery
* Slower	Faster
* larger header	Smaller header
* Maintains connection state	No connection state.

TCP examples => HTTP, HTTPS, FTP, Email

UDP example => DNS, VOIP, Online gaming, video streaming.

② Encapsulation and Network Layer.

(a) Encapsulation and Decapsulation

• Encapsulation:

Encapsulation is the process of adding headers and trailers to data at each layer before transmission.

-> Steps:

① Application layer - Data

② Transport layer - Segment.

③ Network layer - Packet

④ Data Link layer - Frame

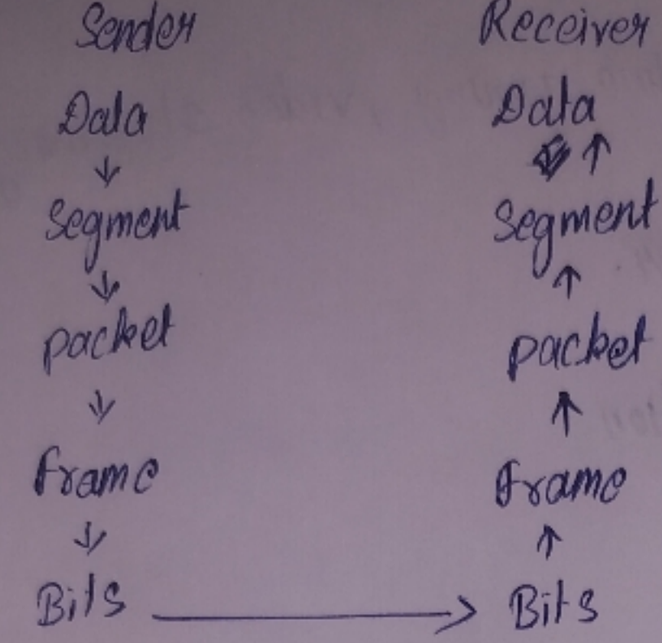
⑤ Physical layer - Bits.

• Decapsulation:

Decapsulation is the process of removing headers and trailers at the destination to obtain the original data.

③

Diagram:



Router function:
- Receives the frame.
- Removes the Data Link header.
- Check the destination IP address.
- Selects the next route.

(b) Functions of Network Layer:
- Routing : Selects the best path.
- IP Addressing : Identifies Sourse and destination
- Routing Table : Helps routers forward packets
- Fragmentation : Divides large packets into smaller parts when acquired.
- TTL : Prevents packets from circulating endlessly.

IP Address vs Mac Address:

IP Address	Mac Address
☆ logical address	physical address
☆ Used at Network layer	used at Data - link layer.
☆ Normally remains same end - to-end	changes at each drop.

③ IP Addressing and Subnetting

(a) Limitations of classful Addressing and CIDR

Limitations of classful Addressing:

1. Fixed network and host boundaries.
2. Wastage of IP address
3. IPV4 address exhaustion.
4. Inflexible Subnetting
5. Large routing tables.

CIDR (classless inter-domain routing):

CIDR uses variable-length prefixes to allocate IP addresses efficiently.

Advantages:

- Reduces IP address wastage
- Supports flexible subnetting
- Improve address allocation.
- Reduces routing table size.

Example: 192.168.0.0/24
192.168.1.0/24
192.168.2.0/24
192.168.3.0/24

Aggregated Network: 192.168.0.0/22.

(b) Subnetting problem

Given:

* IP Address : 192 . 168 . 25 . 130
* Subnet Mask : 255 . 255 . 255 . 192
* CIDR : /26.

% Network ID:

Block size = 252 - 192 = 64.

Subnets:

- 192 . 168 . 25 . 0
- 192 . 168 . 25 . 64
- 192 . 168 . 25 . 128
- 192 . 168 . 25 . 192

Given IP belongs to the third Subnet.

Network ID : 192 . 168 . 25 . 128 .

2. Broadcast Address :

192 . 168 . 25 . 191

3. Usable Host Range :

First Host : 192 . 168 . 25 . 129

Last Host : 192 . 168 . 25 . 190

4. Total Hosts :

$2^6 - 2 = 64 - 2$

$= 62.$

④ Subnetting of 172.16.0.0/16:

Given:

Network Address = 172.16.0.0/16

Required Subnets = 6.

(a) Bits Borrowed and Subnet Mask:

Formula:

$2^n \geq$ No. of subnets.

$2^3 = 8 \geq 6$.

Therefore, borrowed bits = 3

New Prefix:

16 + 3 = 119.

Subnet Mask:

225.255.224.0.

(b) Number of ~~8~~ Subnets and Hosts:

Total Subnets:

$2^3 = 8$

Remaining host bits:

32 - 19 = 13

Usable Hosts:

~~2^{13}~~ $2^{13} - 2$

$= 8192 - 2 = 8190$ hosts.

⑤ Station n :

(c) First Two Subnets

Block size :

256 - 224 = 32

Details	First Subnet	Second Subnet
Network ID	172.16.0.0	172.16.32.0
First Host	172.16.0.1	172.16.32.1
Last Host	172.16.31.254	172.16.63.254
Broadcast.	172.16.31.255	172.16.63.255

Final Answer :

- Borrowed bits = 3
- Subnet mask = 255.255.224.0
- Total Subnets = 8
- Usable hosts = 8190
- Unused Subnets = 2
- Block size = 32.

⑤ Static Routing and Dynamic Routing:

(a) Difference Between static and Dynamic Routing:

Static Routing	Dynamic Routing.
* Manually configured	Automatically configured.
* No automatic updates	updates routes automatically.
* Less adaptable	More adaptable
* No convergence process	Require convergence.
* Suitable for small network	Suitable for large networks.
* Low bandwith usage	uses bandwith for updates.
* less CPU usage	More CPU usage.

Dynamic Routing protocols:

1. RIP - Routing information Protocal.
2. OSPF - Open shortest Path first
3. EIGRP - Enhanced Interior Gateway Routing Protocol.

Static Routing is preferred for:

- ☆ Small networks
- ☆ Single exit path networks.
- ☆ Default routes to ISP.
- ☆ Networks requiring low ref overhead and control.

(b) Static Route configuration:

- Destination Network : 192.168.2.0
- Subnet Mask : 255.255.255.0
- Next HOP : 10.0.0.2

CLI commands:

Router > enable

Router # configure terminal

Router(config) # ip route 192.168.2.0 255.255.255.0 10.0.02

Router (config) # exit

Router # copy running-config startup-config.

Verification: Router # show ip route.

Connectivity Test: Router # ping 192.168.2.1